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Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
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Comments:			
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	EDITOR's NOTE (12/14/2015): Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX14AB34G." Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. The production of innovative materials and the optimization of both properties and processing routes go through an accurate prediction and modeling of the dynamics of microstructures formation. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity experiments provide		

Task Description:	a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. The main goal of this project is to develop state-of-the-art computational tools to model dendritic microstructures formation under diffusive growth conditions and to establish benchmark comparisons with experimental observations under microgravity conditions. Those comparisons have the dual purpose to obtain new fundamental insights into microstructure formation and to validate computational models that can ultimately be used in an industrial setting to guide the design of structural alloys and optimize solidification processing conditions. Computational tools include the phase-field method and a new dendritic needle network (DNN) model. The phase-field method resolves the details of the dendrite tip scale and is ideally suited to model cellular and dendritic structures in dilute alloys on a scale comparable to the primary spacing. In the framework of a collaboration between NASA and the French National Space Agency (CNES), phase-field simulations are being used to model directional solidification experiments in a transparent organic succinonitrile–camphor alloy under low gravity in the Device for the Study of Critical Liquids and Crystallization (DECLIC) on-board the International Space Station (ISS). The DNN model in turn is designed to model dendritic microstructure formation on a grain scale that is orders of magnitude larger than the dendrite tip scale. Unlike previous models of grain structure evolution, the DNN approach models explicitly the growth kinetics of all branches of the hierarchical dendritic network (including primary, secondary, and tertiary branches) that interact diffusively via the solutal field. Hence, this method is ideally suited to investigate complex mechanisms of dendritic microstructure formation including the competitive growth of misoriented columnar grains and the columnar-to-equiaxed transition. This transition is of practical relevance for the casting industry. It is being investigated experimentally aboard the ISS in collaboration with our European partners in the framework of the Columnar-to-Equiaxed Transition in SOLidification Processing (CETSOL) project. Simulation results are critically important to interpret the results of those experiments in both metallic and transparent organic alloys and to guide future experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to lightweight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	During the current year, the following progress was achieved: • CETSOL project o We obtained a better understanding of the CET observed in experimental samples by applying the recently developed 3D extension of the DNN model. The DNN simulations used as input Batch 1 and Batch 2a temperature data to simulate the CET in a thin 3D geometry. The results were compared quantitatively to the measurements of the CET in refined Al-7 wt.% Si alloys solidified in microgravity under different thermal conditions. The DNN modeling predictions for the positions and spatial extent of the CET were found to be in reasonably good quantitative agreement with experimental measurements in two Batch 1 flight samples (FM1 and FM5) that exhibit sharp or progressive CET depending on the thermal conditions. • DECLIC project o We carried out three-dimensional phase-field simulations to investigate the time and initial wavelength of morphological instability of the planar interface that occurs during the transient recoil of the interface. The predicted time and wavelength of instability are in good overall quantitative agreement with experimental measurements and decrease monotonously with the external pulling velocity. The simulation results reveal that an accurate description of the temperature field, which includes latent heat rejection at the interface and diffusion inside the sample and crucible, is essential to predict accurately the time of instability. This time is incorrectly predicted to be much shorter than experimentally observed using the standard frozen temperature approximation, which assumes a linear temperature profile. Those results improve our theoretical understanding of the morphological instability of the planar interface during directional solidification of binary alloys and our ability to predict this instability using accurate numerical models. o We pursued our theoretical investigation of the dynamical selection of the primary pacing during the growth of misoriented crystals by our recently discovered source-sink mechanism. We further developed and tested our analytical model to predict the propagation of the front separating the region of large spacing near the divergent sample boundary, where new cells are created by tertiary branching (source), from the region of small spacing near the convergent boundary, where cell elimination occurs (sink). This model, which is based on a nonlinear advection equation, predicts that this front moves at about one third of the lateral drift velocity that is geometrically imposed by the crystal misorientation. This prediction is an excellent quantitative agreement with phase-field simulations and thin-sample terrestrial experiments. o We studied the grain boundary instability at the convergent grain boundary of a 3D array. The experimental observations reveal that, when two grains drift towards each other due to their crystalline orientations, one grain can penetrate into the other nearby grain. The PF simulations reveal that a grain with a low misorientation (= 10°) can penetrate into the other well-orientated grain. This penetration process leads to the grain boundary instability. The next steps in the CETSOL project include: • To continue to use the dendritic needle network (DNN) numerical model to characterize the columnar-to-equiaxed

	transition during solidification of AlSi7 alloys microgravity experiments performed in the MSL-SQF furnace on ISS. Modeling in this next phase will focus on characterizing the fully three-dimensional dendritic microstructure and larger scale grain structure of the CETSOL Batch2a samples. Comparisons of modeling predictions will be made with three-dimensional reproduction of the dendritic microstructure and grain structure of relevant parts of the samples provided by the partnering team at Northwestern University. The comparison with experimental data will be used for verification and improvement of the model. • To contribute to the numerical modeling part of the CETSOL-6 proposal that is a direct extension of the ongoing CETSOL-5 project. Numerical modeling in the CETSOL-6 project will focus on investigating the columnar-to-equiaxed transition in Batch3a samples that use different alloys (AlCu instead of AlSi) with different compositions and equiaxed solidification in microgravity experiments using binary transparent alloys in the TRANSPARENT ALLOY facility on ISS. These experiments provide in-situ observation of solidification dynamics under diffusive growth conditions and provide further data for validation of numerical models.
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